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TRM-VLA: Temporal-Aware Chain-of-Thought Reasoning and Memorization for Vision-Language-Action Models

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Abstract

Vision-Language-Action (VLA) models have emerged as a powerful paradigm for general robotic manipulation. However, existing approaches typically omit intermediate reasoning steps and directly regress actions, limiting reasoning interpretability and performance in long-horizon or compositional tasks. Although recent studies introduce Chain-of-Thought (CoT) reasoning into VLA models, their effectiveness remains suboptimal due to two key issues: (1) generating a full reasoning trajectory at every timestep introduces substantial redundancy, thereby hinders real-time deployment and (2) reasoning is performed independently, neglecting temporal consistency, which leads to planning conflicts. We propose TRM-VLA, a temporal-aware reasoning and memorization framework that integrates explicit temporal modeling into the VLA reasoning process. TRM-VLA consists of two core components: (1) Keyframe-Triggered Reasoning (KTR), which identifies task progress and performs hierarchical CoT reasoning only at key decision points to reduce redundant inference; and (2) Granularity-adaptable Context Memory (GCM), which dynamically stores and retrieves historical reasoning trajectories to maintain inter-frame coherence and global context. Extensive experiments on LIBERO-90, SIMPLER, and four real-world robotic tasks demonstrate that TRM-VLA achieves state-of-the-art performance while improving reasoning efficiency.

1. Introduction

Recent years have witnessed significant progress in general-purpose robotic manipulation policies [4, 5, 18, 25, 31, 41, 58]. Among these advances, VLA models have emerged as a powerful paradigm by leveraging the multimodal understanding of pre-trained Vision-Language Models (VLMs) to enhance embodied generalization [2, 6, 14, 18, 19, 29, 58]. By fine-tuning on robot interaction data, VLA models inherit open-world reasoning capabilities from VLMs [3, 16], mapping visual observations and language instructions directly to action spaces for execution. Despite promising results, most existing approaches suffer from the erosion of pre-trained knowledge from the VLM [56] during fine-tuning and lack explicit reasoning processes [18, 22, 58], directly regressing actions from inputs. This omission undermines manipulation interpretability and limits their performance on long-horizon, complex tasks that require structured planning.

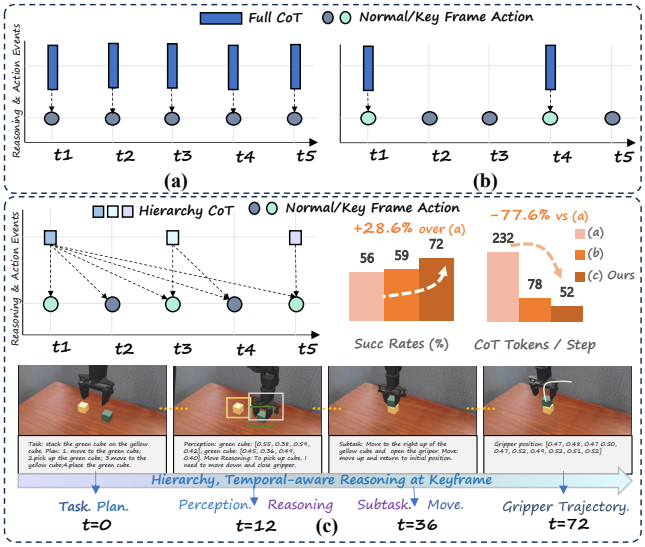


Figure 1. Different reasoning patterns of VLAs. (a) generate full CoT at every frame. (b) generate CoT at fixed intervals without memory. (c) generate hierarchy CoT at critical frames, and maintain a dynamic memory buffer for execution, thus reducing redundant CoT reasonings tokens while improving the success rates.

Inspired by the chain-of-thought (CoT) reasoning [1, 44] in large language models (LLMs), wherein complex problems are decomposed into intermediate steps through prompting, thereby enhancing reasoning performance. In robotics, integrating CoT holds promise for grounding reasoning in textual instructions, visual observations, and actions. Prior

reasoning-augmented works have explored various strategies: some incorporate embodied-relevant primitives such as keypoints, bounding boxes, or trajectories as reasoning intermediates [11, 14, 26, 50]; others leverage action or video generation models or world models to produce goal-conditioned visual plans that guide step-by-step execution [48, 52]. Additionally, several approaches mitigate knowledge erosion in VLA models through co-training on internet-scale language data and robot interaction data [37, 45–47, 56].

However, as shown in Figure 1(a)(b), current reasoning-augmented VLA methods generate a full and independent CoT at every timestep or at fixed intervals, without considering the temporal structure of the task. This design leads to two critical issues: 1) *Redundant reasoning*. Most methods generate a full CoT trajectory at every timestep, even when adjacent frames share nearly identical visual and linguistic contexts. This results in repetitive reasoning, excessive CoT token generation, and substantial computational overhead with diminishing decision gains. 2) *Independent inter-frame reasoning*. Each CoT is produced in isolation, without temporal dependency or memory. The absence of historical reasoning causes contradictory action plans, breaking temporal coherence and degrading performance on long-horizon tasks. For instance, in sequential button pressing(e.g., "press red, then green, then blue"), static visuals prevent the agent from tracking which steps are completed, causing repetition and task failure. This non-Markovian dependency cannot be resolved by such frame-wise reasoning [39].

To address these limitations, we propose **Temporal-aware Reasoning and Memorization** for Vision-Language-Action models, TRM-VLA — a framework that introduces structured, keyframe-trigger reasoning and explicit context-aware memory mechanisms into the VLA pipeline.

(1) *Keyframe-Triggered Reasoning (KTR)*. KTR performs hierarchical CoT reasoning only at key decision points (e.g., when the robot grasps the object) and is informed by task progress, thereby reducing redundant reasoning and token generation while preserving interpretability.

(2) *Granularity-adaptable Context Memory (GCM)*. To overcome the lack of temporal consistency and memory, GCM maintains a dynamic memory buffer of past critical reasoning steps, which are selectively retrieved through cross-attention to inform current decisions. By integrating historical reasoning into the policy state, GCM enforces temporal coherence, allowing the model to maintain consistent plans and to leverage memory to track task progress.

As illustrated in Figure 1, unlike Pattern (a) [47, 50] (full CoT at every timestep) or Pattern (b) [10, 27] (CoT generated only at fixed intervals without memory), our approach combines hierarchy, keyframe-triggered reasoning with dynamic, multi-granularity memory. This breaks the implicit Markovian assumption of prior methods, endowing TRM-VLA with reasoning efficiency and temporal awareness essential for complex, real-world manipulation tasks.

Our TRM-VLA framework builds upon a state-of-the-art multimodal foundation model [16, 18] and a Diffusion Transformer [22, 34] action policy. We adopt a dual-system architecture [6] inspired by cognitive science: the foundation model acts as System 2, responsible for slow, deliberate CoT generation, while the Diffusion Transformer [9] serves as System 1, executing fast, reflexive action prediction. The model is pretrained on the Open X-Embodiment dataset [32] and subsequently fine-tuned on task-specific robot interaction data for deployment. We annotate existing embodied datasets with multi-level temporal labels and employ causal attention with next-token prediction as training objective, enabling the VLM to learn to implicitly identify task progress and generate reasoning at appropriate levels of abstraction, from strategic planning to concrete motor commands.

Extensive experiments on LIBERO-90, SIMPLER, and four real-world manipulation tasks demonstrate that TRM-VLA not only achieves state-of-the-art performance (72.9% on SIMPLER [23]) but also reduces CoT token generation by 4× compared to existing reasoning-enabled VLA models.

- We propose the TRM-VLA framework, which integrates temporal-aware reasoning and memorization to address the inference redundancy and inconsistency with frame-wise CoT generation for effective robotic manipulation.
- We introduce keyframe-triggered reasoning (KTR) and granularity-adaptable context memory (GCM), which guarantee temporally-coherent planning by incorporating hierarchical CoT and dynamic memory retrieval.
- We conduct comprehensive evaluation in real and simulated settings, demonstrating superior performance and efficiency across a diverse set of manipulation tasks.

2. Related Works

2.1. Generalist Robot Manipulation Policies

Recent work has demonstrated that scaling robot manipulation policies across diverse environments and tasks significantly improves their multi-task and cross-scenario generalization capabilities [2, 17, 32, 43]. Training such generalist policies typically requires scalable architectures capable of handling heterogeneous, large-scale, cross-embodiment robot interaction data. Vision-Language-Action models [14, 18, 22, 29, 30, 32, 36, 41, 45, 58] offer a promising paradigm: they build upon pre-trained Vision-Language Models and perform post-training on robot data. This approach effectively transfers the rich multimodal understanding of VLMs to embodied control, circumventing the data-hungry nature of training from scratch. Subsequent research has explored multiple directions to enhance VLA performance, including: Action generation architectures, such as Diffusion-based Action Experts for continuous control [4, 22, 45]; Dual-system designs that separate high-level reasoning from low-level execution [6, 40]; Multimodal data composition strate-

gies, incorporating tactile, 3D, or instruction-augmented inputs [13, 14, 47, 53, 56] and reinforcement learning fine-tuning to refine policies post-supervised training [12, 21, 49]. While chain-of-thought (CoT) reasoning has substantially advanced LLM-based reasoning, its integration into VLA models has brought only marginal improvements. Crucially, most existing methods neglect the temporal structure of reasoning inherent to robotic manipulation—where decisions are non-Markovian and depend on historical context beyond the current observation. In this work, we bridge this gap by introducing temporal-aware hierarchical reasoning with persistent memory as an intermediate representation prior to action generation, enabling temporally consistent decision-making in long-horizon and complex manipulation tasks.

2.2. Reasoning in Vision-Language-Action Models

Existing reasoning-enabled VLA approaches [8, 14, 50] can be broadly categorized into three lines of work: (1) Primitive-based reasoning. These methods use embodied-relevant primitives such as subgoals, trajectories, object bounding boxes, or keypoints as explicit reasoning intermediates [11, 14, 20, 26, 52, 57]. By grounding reasoning in task-relevant visual or geometric cues, they improve policy focus and performance. (2) World model or video-generation-augmented reasoning. Some works employ world models or video generation models to produce goal-conditioned visual plans (e.g., goal images) that guide step-by-step execution [7, 52]. Others learn latent action representations from human demonstration videos and transfer them as priors to robots [2, 48]. (3) Co-training with internet-scale and embodied data. To mitigate language forgetting during robot fine-tuning, several approaches jointly train on internet-scale multimodal data (e.g., image captions, VQA pairs) and robot interaction data [14, 37, 46, 47, 56]. In contrast, we propose a temporal-aware reasoning and memory framework for VLAs: reasoning is triggered only at key decision points, and past reasoning traces are explicitly stored and retrieved to inform current decisions. This design better aligns with the memory-dependent nature of real-world robotic tasks.

3. Method

We present our Temporal-aware Reasoning and Memorization framework for VLAs, designed to enhance the temporal reasoning and memory in robotic manipulation. We first formalize standard and reasoning-augmented VLA models in Sec. 3.1, followed by an overview of our system architecture in Sec. 3.2 as shown in Fig. 2. We then detail the two core modules, Keyframe-Triggered Reasoning in Sec. 3.3 and Granularity-adaptable Context Memory in Sec. 3.4.

3.1. Preliminaries

Standard VLA aims to map visual observations o_t and language instructions l_t into robot actions a_t at timestep t:

$$a_t \sim P_\theta(a_t \mid o_t, l_t) \tag{1}$$

where $a_t \in \mathbb{R}^d$ typically represents a 7-DoF vector (e.g., $[\Delta x, \Delta\beta, \text{gripper}]$), depending on the robot embodiment, where $\Delta x, \Delta\beta, \text{gripper}$ denote the relative translation, rotation and gripper state of the end effector, and θ is the parameters of the network. While effective for short-horizon tasks, this direct mapping discards intermediate reasoning, leading to poor interpretability and degraded performance on complex, multi-stage robotic manipulation.

Reasoning-augmented VLA introduces an intermediate reasoning trace r_t, such as textual planning or visual subgoals, to bridge the gap between perception and action:

$$a_t \sim P_\theta(a_t \mid r_t, o_t, l_t), \qquad r_t \sim P_\theta(r_t \mid o_t, l_t) \tag{2}$$

However, these methods typically generate r_t independently at every timestep, ignoring temporal structure. This leads to (1) redundant computation due to near-identical reasoning across consecutive frames, and (2) inconsistent subgoal sequencing that undermines long-horizon coherence.

Temporal-Aware Reasoning Formulation for VLA. We explicitly model temporal dependencies by conditioning action prediction on a memory-augmented reasoning state m_t, integrating both current and historical reasoning contexts.

$$a_t \sim P_\theta(a_t \mid m_t, o_t, l_t) \tag{3}$$

$$m_t \sim P_\phi(m_t \mid r_1^h, \ldots, r_t^h) \tag{4}$$

where $r_t^h \sim P_\theta(r_t^h \mid o_t, l_t)$ denotes the hierarchy, keyframe-triggered CoT reasonings produced by our Keyframe-Triggered Reasoning module (KTR). Then Granularity-adaptable Context Memory (GCM) module (P_ϕ) dynamically aggregates past reasoning traces $\{r_1^h, \ldots, r_t^h\}$ into a compact, temporally consistent representation m_t. By generating keyframe-triggered reasonings and propagating memory-based context across timesteps, TRM-VLA establishes a coherent temporal reasoning process. This design enables the model to maintain global reasoning context while executing efficient, consistent control across long-horizon and complex tasks.

3.2. TRM-VLA Framework

As shown in Fig. 2, TRM-VLA builds upon CogACT [22], a strong baseline that combines a multimodal foundation model with a diffusion-based action policy inspired by cognitive architectures [6]. Our framework follows a dual-system design: a deliberative System 2 (VLM-based reasoning) and a reactive System 1 (diffusion action expert).

The visual encoder transforms the input image into visual tokens $V = \{v_1, v_2, \ldots, v_{N_v}\}$. It consists of two complementary vision transformers pretrained on large-scale internet imagery: DINOv2 [33] for capturing spatially localized features, and SigLIP [51] for encoding semantic-level representations. Together, they provide a comprehensive visual understanding for downstream reasoning.

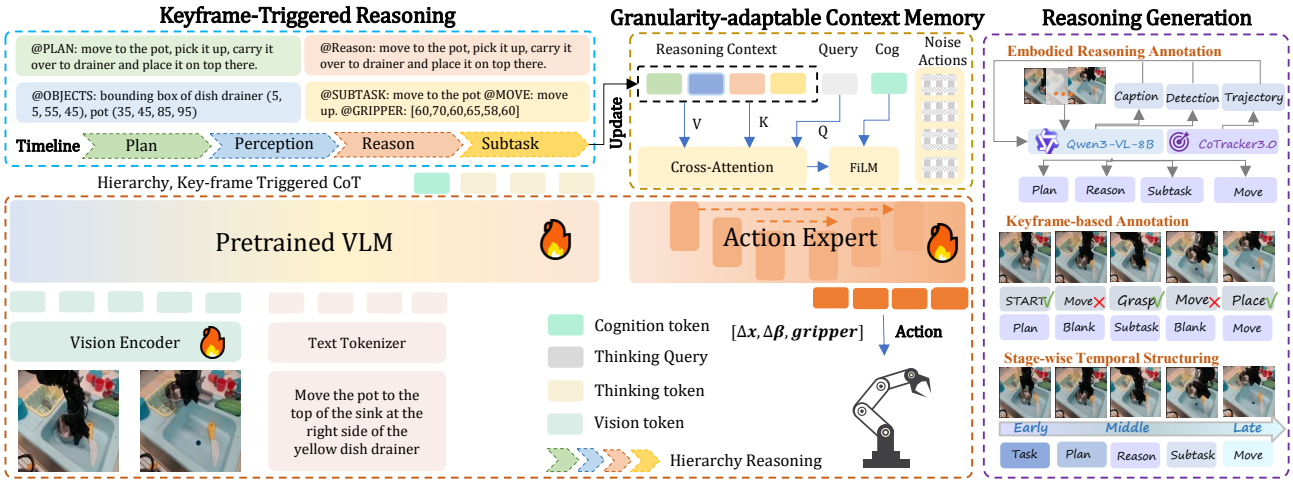


Figure 2. Overview of the proposed TRM-VLA (Sec. 3.2). (a) System design: VLM backbone for reasoning and diffusion expert for execution. (b) Keyframe-Triggered Reasoning (Sec. 3.3) performs hierarchical reasoning only at critical frames supervised by temporal reasoning data. (c) Granularity-adaptable Context Memory (Sec. 3.4) maintains a dynamic memory buffer to enhance inter-frame reasoning coherence.

The language encoder adopts LLaMA-2 [42], which jointly processes the tokenized instruction $L = \{l_1, l_2, \ldots, l_{N_l}\}$, visual tokens V, and a learnable cognition token c through a multimodal transformer. During training, the model is trained on embodied reasoning data annotated with multi-level keyframes and temporal stages, enabling it to learn when and how to reason. At inference time t, it adaptively outputs: 1) a hierarchy reasoning r_t^h through Keyframe-Triggered Reasoning module whenever t corresponds to a keyframe (and no reasoning is generated otherwise), and 2) a cognition feature f_t corresponding to the cognition token c, which encodes the multimodal representation of the current scenario.

To ensure temporal consistency and contextual reasoning, Granularity-adaptable Context Memory (GCM) maintains a dynamic memory buffer of past critical reasoning features and constructs a memory-augmented state m_t. The model integrates m_t with f_t via a FiLM-based [35] fusion network, effectively propagating temporal dependencies across frames and facilitating consistent reasoning over time. Finally, the action expert—a Diffusion Transformer (DiT) [34]—takes the fused cognition feature as a conditioning input. Given Gaussian noise and f_t the DiT performs iterative denoising to generate an action chunk $A = (a_1, a_2, \ldots, a_{N_a})$. This diffusion-based decoding yields smooth, temporally coherent action sequences, crucial for stable real-world execution.

3.3. Keyframe-Triggered Reasoning

Our core insight is that effective reasoning in embodied tasks should "think appropriately" at each timestep, generating reasoning most relevant to the current scene (Eq. 3)—and to "think temporally" by integrating historical reasoning across time (Eq. 4). Together, these two capabilities strengthen the model's temporal reasoning and allow it to perform embodied decision-making with long-horizon consistency. KTR achieves this by producing sparse, hierarchical CoT only at critical decision points, thereby aligning reasoning effort with task dynamics. As shown in Fig. 2, KTR is trained on embodied reasoning data augmented with fine-grained temporal annotations, as described below.

Embodied Reasoning Annotation. We extend existing robot manipulation datasets with structured CoT annotations, following the ECoT protocol [50]. Each reasoning trace is decomposed into semantically distinct components: $tag \in \{task, plan, perception, subtask\ reasoning, subtask, move\ reasoning, move, gripper\ position\}$, which collectively capture high-level task decomposition and low-level, perception-grounded action justification. At timestep t , this yields a structured label $R_t = \{r_t^{tag}\}$. To improve the quality and temporal consistency of annotations, we introduce three key enhancements: (1) a temporal tracking model [15] for smooth gripper and object trajectory alignment; (2) a unified VLM [3] for joint object detection, captioning, and reasoning to ensure semantic coherence; and (3) an optimized annotation pipeline for improved inter-frame consistency and efficiency. Details can be found in the Appendix.

Keyframe-Based Annotation. Frame-wise annotation introduces severe redundancy: high-level tags (e.g., $plan$) remain static for entire episodes, while low-level tags (e.g., $gripper\ position$) change rapidly. To mitigate this, we introduce *hierarchical keyframe indicators* that mark meaningful transitions in reasoning structure. Specifically, we define binary flags for three reasoning granularities:

$$b_t^\tau = \begin{cases} 1, & \text{if } r_t^\tau \neq r_{t-1}^\tau, \\ 0, & \text{otherwise,} \end{cases} \quad \tau \in \{\text{per, s, m}\}, \tag{5}$$

where τ = per(perception), s(subtask), m(move) denote levels of temporal granularity. This hierarchical scheme produces three types of keyframes: b_t^{per} stands for signal significant scene or object-state changes; b_t^{s} marks transitions between distinct subgoals; b_t^{m} captures fine-grained motion or gripper adjustments. These keyframes serve as supervision signals for when reasoning should be updated.

Stage-Wise Temporal Structuring. To further model temporal dependencies and reduce redundant reasoning, we segment each task episode into three temporal phases: an early stage for high-level planning (task, plan, perception, subtask/move reasoning), a middle stage for perception-anchored subtask decomposition (activated only at perception keyframes), and a late stage for low-level execution (subtask/move commands and gripper control, triggered by subtask or move keyframes). The temporal-aware hierarchical reasoning set $\mathcal{T}_t$ is then defined as:

$$\mathcal{T}_t = \begin{cases} \{r_t^{\text{t}},\ r_t^{\text{p}},\ r_t^{\text{per}},\ r_t^{\text{sr}},\ r_t^{\text{mr}}\}, & t < t_{\text{e}}, \\ \{r_t^{\text{per}},\ r_t^{\text{sr}},\ r_t^{\text{mr}}\}, & t_{\text{e}} < t < t_{\text{m}} \text{ and } b_t^{\text{per}} = 1, \\ \{r_t^{\text{s}},\ r_t^{\text{m}},\ r_t^{\text{g}}\}, & t_{\text{m}} < t < t_{\text{l}} \text{ and } b_t^{\text{s}} = 1, \\ \{r_t^{\text{m}},\ r_t^{\text{g}}\}, & t_{\text{m}} < t < t_{\text{l}} \text{ and } b_t^{\text{m}} = 1. \end{cases} \tag{6}$$

where we use abbreviated tag names for brevity (e.g., t = $task$, p = $plan$, per = $perception$, sr = $subtask\ reasoning$, mr = $move\ reasoning$, g = $gripper\ position$). Note that reasoning is only generated when a corresponding keyframe is active, ensuring sparsity. We train KTR using a standard next-token prediction objective with causal attention:

$$\mathcal{L}_{\text{KTR}} = -\sum_{S \in \mathcal{D}} \sum_{t}^{T} log\ p(\mathcal{T}_t | o_t, l_t; \theta) \tag{7}$$

where D denotes the annotated dataset and T the episode length. This objective encourages the model to learn *when* to reason (via keyframe-triggered activation) and *what* to reason about (via hierarchical tag prediction). In summary, KTR equips the VLA model with adaptive temporal reasoning: it automatically identifies critical decision points, generates multi-granularity CoT aligned with task structure, and suppresses redundant computation. This enables a smooth transition from abstract planning to concrete execution—mirroring human-like problem solving in embodied tasks.

3.4. Granularity-adaptable Context Memory

While the KTR module enables the model to produce hierarchical reasoning aligned with temporal keyframes, relying solely on the reasoning of the current timestep can be suboptimal. For instance, when the model generates only low-level reasoning at a subtask keyframe, it may overlook high-level planning information that remains relevant throughout the task. To address this limitation, we introduce the GCM module, which maintains a dynamic, multi-granularity memory of past reasoning to form a temporally coherent context C, enabling the model to "reason across time."

Dynamic Reasoning Context. In a task of length T with N_k keyframes $\{k_i\}_{i=1}^{N_k}$ where $0 < k_1 < \cdots < k_{N_k} < T$, the reasoning context at the current keyframe k_c is represented as a dictionary-based memory buffer C_{k_c}, which stores hierarchical reasoning tokens $\{r_{k_i}^{\text{tag}}\}_{i=1}^{c}$ from previously observed keyframes. When the model generates new reasoning components $r_{k_c}^{\text{tag}}$, they are inserted into C_{k_c}, overwriting any existing entries with the same tag.

$$C_{k_c} = C_{k_{c-1}} \cup \{r_{k_c}^{\text{tag}}\}, \tag{8}$$

To account for the temporal stability of different reasoning levels, we assign hierarchical lifetimes: high-level reasoning ($r_k^{\text{task}}, r_k^{\text{plan}}$) is generated mainly in the early stage and preserved in the context until the task ends. Mid-level reasoning ($r_k^{\text{p}}, r_k^{\text{sr}}, r_k^{\text{mr}}$) is retained for a moderate duration, replaced only when new reasoning of the same tag appears. Low-level reasoning ($r_k^{\text{s}}, r_k^{\text{m}}, r_k^{\text{g}}$) has a shorter lifespan, updated frequently to reflect real-time control variations. Compared with per-frame CoT reasoning, GCM provides a memory-efficient alternative that preserves long-range task consistency without redundant computation.

Temporal Reasoning Integration. To inject historical context into action generation, GCM employs a learnable thinking query q_t that retrieves the most relevant reasoning features from C_t. Specifically, we encode all entries in C_t through the vision-language model to obtain contextual embeddings f_{rc}, which serve as keys and values in a cross-attention mechanism. To avoid redundant computation, both C_t and f_{rc} are jointly maintained and updated. This design enables GCM to efficiently reuse past reasoning without repeated forward passes through the VLM, reducing inference cost while preserving temporal coherence and context:

$$f_{\text{att}} = \text{CrossAttn}(q,\ K = f_{\text{rc}},\ V = f_{\text{rc}}), \tag{9}$$

The retrieved feature f_{att} is then fused with the current cognition token feature f_c (from the VLM) via a Feature-wise Linear Modulation (FiLM) network [35]:

$$f_t = \text{FiLM}(f_c,\ f_{\text{att}}), \tag{10}$$

yielding a fused feature f_t that encodes both immediate perception and historical reasoning. This context-enriched representation serves as the conditioning signal for the downstream diffusion-based action policy.

Action Diffusion and Training Objective. The fused feature f_t is passed to a diffusion-based action expert [34], which iteratively denoises a latent action sequence to produce a smooth action chunk $A = (a_1, a_2, \ldots, a_{N_a})$. Following standard diffusion policy training, we minimize the mean squared error between predicted and ground-truth noise:

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1), i} \left| \hat{\epsilon}^i - \epsilon \right|_2 \tag{11}$$

where $\hat{\epsilon}^i$ is the noise prediction at denoising step i for a noisy action sequence $(a_t^i, a_{t+1}^i, \ldots, a_{t+N}^i)$. This learning objective encourages the policy to generate actions that are not only physically plausible but also semantically grounded in temporally consistent reasoning.

4. Experiments

4.1. Datasets and Benchmarks

SIMPLER Benchmark. We evaluate on SIMPLER [23], a high-fidelity simulation suite that closely matches real-world robot dynamics and visual appearance. A key feature of SIMPLER is its real-to-sim evaluation protocol: policies trained on real-world data are tested in simulation, with strong empirical correlation to physical-world performance under distribution shifts. This makes SIMPLER ideal for assessing reasoning-intensive methods, as it preserves the semantic and temporal structure of manipulation tasks while enabling scalable, reproducible evaluation.

LIBERO Benchmark. We further test on LIBERO-90 [28], a challenging long-horizon benchmark comprising 90 language-conditioned tasks across 10 household domains. Tasks require fine-grained object interaction, multi-stage planning, and precise control—making it a rigorous testbed for temporal reasoning and memory.

Real-World Benchmark. We collect 410 expert demonstration trajectories on an AIRBOT Player robot across four carefully designed tasks that probe distinct reasoning capabilities: (1) *Scoop Washers with Shovel*: The robot uses a non-end-effector tool (shovel) to scoop small metal washers, testing generalization to tool-based manipulation; (2) *Clean Table*: The robot clears diverse objects (bear, banana, cup, etc.) into a bin, evaluating long-horizon planning and object-agnostic execution; (3) *Push Buttons in Order*: The robot presses three colored buttons in a specified sequence, explicitly testing temporal memory and sequential reasoning; (4) *Place Toy Bear into Box*: The robot grasps a deformable object from varied poses and places it into a container, assessing robustness to object shape and pose variations. Together, these tasks span tool use, memory-dependent sequencing, sustained planning, and precise manipulation—core challenges addressed by our framework.

Embodied Reasoning Datasets. For the SIMPLER-WidowX benchmark, we build upon the BridgeData V2 [43] dataset and adopt the CoT annotations generated by ECoT [50]. For LIBERO-90 [28], we use the reasoning annotations from ECoT-Lite [8], a streamlined variant of ECoT tailored to the LIBERO task suite. For real-world experiments, we employ the Qwen3-VL [3] multimodal language model to automatically produce embodied CoT labels from video-language-action sequences. Our annotation pipeline simplifies and consolidates the original protocol resulting in higher annotation quality and efficiency. [1]

4.2. Quantitative Results

Simulated Evaluation. We evaluate TRM-VLA on the SIMPLER-Bridge benchmark [23], which provides high-fidelity simulated environments. For each task, we report Grasp Rate (percentage of trials with successful object contact) and Success Rate (SR) (full task completion). As shown in Table 2, TRM-VLA achieves state-of-the-art performance across all tasks, significantly outperforming prior VLA and generalist policy models. Notably, TRM-VLA attains a 72.9% average success rate, surpassing the baseline method [22] by +21.6%. This substantial gain underscores the effectiveness of our temporal-aware reasoning and memory framework in long-horizon manipulation.

We further evaluate TRM-VLA on the LIBERO-90 benchmark [28]. As shown in Table 1, TRM-VLA achieves a new state-of-the-art success rate of 94.8%, surpassing all prior methods by a significant margin. Notably, TRM-VLA outperforms the baseline model, CogACT-Base (88.4%), by +6.4%, and exceeds strong contemporaries like π_0-Fast (83.1%) and UniACT (73.0%). This demonstrates that our framework excels at complex, multi-stage tasks requiring sustained planning and temporal awareness.

Real-World Evaluation. We evaluate TRM-VLA on four challenging real-world manipulation tasks designed to probe distinct aspects of embodied reasoning: Place Toy Bear into Box (precision grasping), Push Buttons in Order (memory-dependent sequencing), Clean Table (long-horizon planning), and Scoop Washers using Shovel (tool use). As shown in Tab. 3, TRM-VLA achieves the highest success rate across all four tasks, outperforming strong baselines.

On Push Buttons in Order, which demands explicit memory of the button sequence, TRM-VLA achieves a 67% success rate, significantly surpassing CogACT-ECoT's 45%. This gain underscores the effectiveness of our GCM module in preserving and leveraging historical reasoning traces. On Clean Table, a multi-step, long-horizon task involving object clearing, TRM-VLA attains a 72% success rate, compared to 41% for CogACT-ECoT. This highlights the benefit of our hierarchical, coarse-to-fine reasoning via KTR, which enables sustained planning over extended task horizons. Furthermore, TRM-VLA exhibits superior inference efficiency: it generates only 4.3 CoT tokens per step on average, a dramatic reduction compared to ECoT (26.8). This efficiency stems from our key design principle: triggering reasoning only at critical frames rather than generating full CoT at every timestep. This reduces computation and mitigates context pollution, allowing the policy to focus on essential decision points. As shown in Fig. 4, our method accurately identifies keyframes, and the generated hierarchical CoT

[1] More details about **dataset**, **metrics**, **implementation details**, holistic ablations, visualizations, and related works are available in the **Appendix**.

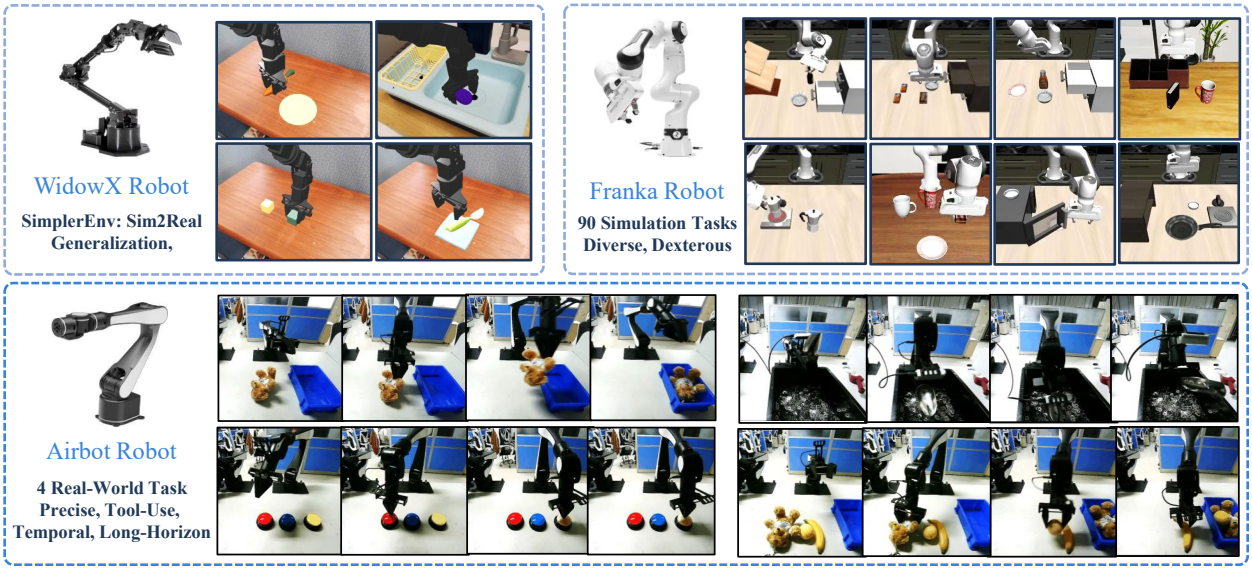


Figure 3. Cross-task, cross-embodiment evaluation environments: Simpler-Env [23] with WidowX robot, LIBERO-90 [28] with Franka robot, and real-world tasks based on AIRBOT Player robot, focusing on manipulation accuracy, tool-using, temporal memory, and long-horizon.

Model	Publication	Put Spoon		Put Carrot		Stack Cube		Put Eggplant		Avg. SR(%) ↑
		Grasp(%)	SR(%)	Grasp(%)	SR(%)	Grasp(%)	SR(%)	Grasp(%)	SR(%)	
RT-1-X [5]	ICRA 2024	16.7	0.0	20.8	4.2	8.3	0.0	0.0	0.0	1.1
OpenVLA [18]	CoRL 2025	4.1	4.2	33.3	0.0	12.5	0.0	8.3	4.1	1.0
SpatialVLA [38]	Arxiv 2024	20.8	16.7	29.2	25.0	62.5	29.2	**100.0**	**100.0**	42.7
Magma [46]	CVPR 2025	-	37.5	-	29.2	-	20.8	-	91.7	44.8
π_0-Fast [36]	Arxiv 2025	-	29.1	-	21.9	-	10.8	-	66.6	32.1
Octo-Base [31]	ICRA 2024	34.7	12.5	52.8	8.3	31.9	0.0	66.7	43.1	17.5
TraceVLA [55]	ICLR 2025	-	12.5	-	16.6	-	16.6	-	65.0	27.7
RoboVLM [24]	Arxiv 2025	70.8	45.8	33.3	20.8	54.2	4.2	91.7	79.2	37.5
CogACT-Base [22]	Arxiv 2025	79.2	71.7	66.7	50.8	54.2	15.0	75.0	67.5	51.3
π_0 [4]	CoRL 2025	-	**83.8**	-	52.5	-	**52.5**	-	87.9	69.2
TRM-VLA	——	**91.7**$^{\pm.08}$	83.3$^{\pm.08}$	**79.2**$^{\pm.04}$	**75.0**$^{\pm.04}$	**79.2**$^{\pm.04}$	41.7$^{\pm.04}$	91.7$^{\pm.08}$	91.7$^{\pm.08}$	**72.9**$^{\pm.04}$ (+21.6)

Table 1. Quantitative comparison with SOTA methods on SimplerEnv-Bridge benchmark. 'SR' indicates the success rates.

Method	Publication	Success Rate(%)
SpatialVLA [38]	Arxiv 2025	46.2
OpenVLA [18]	CoRL 2025	73.5
UniACT [54]	CVPR 2025	73.0
CogACT-Base [22]	Arxiv 2025	88.4
CogACT-ECoT [50]	CoRL 2025	92.1
π_0-Fast [36]	Arxiv 2025	83.1
TRM-VLA	——	**94.8**(+6.4)

Table 2. Comparison with SOTA methods on LIBERO-90.

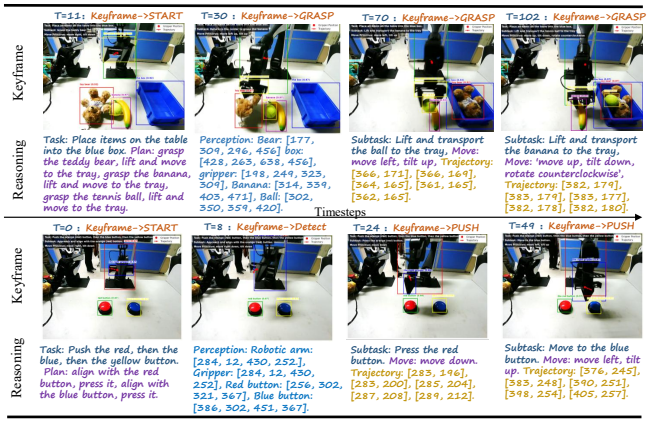


Figure 4. Qualitative Results on real-world evaluation. The TRM-VLA model adaptively triggers reasoning at critical frames.

aligns well with the actual actions. Refer to the appendix.

Out-of-Distribution Evaluation. We evaluate robustness under real-world distribution shifts, including changes in lighting, background, distractors, spatial layout, and camera pose. As shown in Fig. 5, TRM-VLA consistently outperforms the CogACT baseline [22]. The bar charts quantify the drop in success count ($-\Delta$ success count) under each shift, where a lower value indicates better robustness. In "Push Buttons in Order" task, which requires memory and

Model	Place Toy Bear Progress(%)	Place Toy Bear Success	Push Buttons in Order Progress(%)	Push Buttons in Order Success	Clean the Table Progress(%)	Clean the Table Success	Scoop Washers Progress(%)	Scoop Washers Success	Avg. SR(%) ↑	Avg. Tokens ↓
OpenVLA [18]	0.40	6/20	0.21	3/20	0.15	2/20	0.30	6/20	21.0	7.0
CogACT-Base [22]	0.56	10/20	0.51	9/20	0.37	7/20	0.70	14/20	50.0	1.0
ECoT [50]	0.42	7/20	0.27	5/20	0.18	3/20	0.35	7/20	28.0	26.8
CogACT-ECoT [50]	0.52	8/20	0.45	8/20	0.41	6/20	0.60	12/20	44.0	26.3
TRM-VLA	0.77	15/20	0.67	12/20	0.72	11/20	0.85	17/20	69.0	4.3

Table 3. Quantitative comparison with SOTA methods on real-world tasks. 'SR' indicates the success rates. 'Tokens' denotes the average number of generated CoT tokens per step, and 'Progress' is the average completion progress of the task.

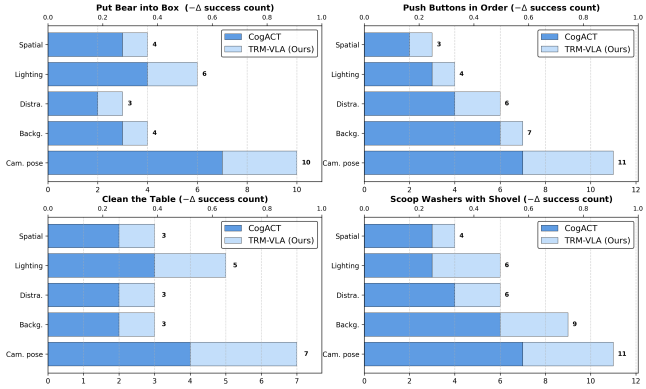


Figure 5. Qualitative Results on generalization evaluation.

sequential reasoning, CogACT's success drops by 7 trials under camera pose variation, while TRM-VLA's drop is only 4. Similarly, on "Clean the Table," a long-horizon task, TRM-VLA maintains higher stability under lighting and background changes. This enhanced robustness arises from temporal memory aggregation: by fusing past reasoning traces, TRM-VLA maintains a consistent understanding of the task goal even when perceptual cues vary. It thus focuses on semantic and procedural invariants rather than transient visual features, enabling stronger generalization.

4.3. Ablation Study

Inference Efficiency. Since all compared models share the same backbone [16], generated CoT tokens per timestep serve as a direct proxy for computational cost. As shown in Tab. 1, our TRM-VLA achieves superior efficiency: it generates only 4.3 tokens per step, a dramatic reduction compared to prior reasoning-augmented models. For instance, while ECoT (the reasoning version of OpenVLA) improves the success rate over its base model, it incurs a 3.8x increase in token generation (7.0 vs. 26.8). Similarly, CogACT-ECoT builds upon CogACT-Base but increases the tokens from 1.0 to 26.3. In contrast, TRM-VLA achieves a trade-off between performance and efficiency. This efficiency stems from our design principle of triggering reasoning only at critical frames, thereby avoiding redundant computation.

Effectiveness of KTR and GCM. As shown in Tab. 4 and Tab. 5, both KTR and GCM modules are essential. On the SIMPLER, removing either KTR (ID a) or GCM (ID c)

ID	KTR KBA	KTR STS	GCM DRC	GCM TRI	Put Spoon	Put Carrot	Stack Cube	Put Eggplant	Avg. SR
(a)	✗	✗	✗	✗	0.67	0.63	0.21	0.67	0.54
(b)	✗	✓	✗	✗	0.75	0.50	0.33	0.83	0.60
(c)	✓	✓	✗	✗	0.83	0.58	0.25	0.92	0.65
(d)	✓	✓	✗	✓	0.92	0.63	0.29	0.92	0.69
(e)	✓	✓	✓	✗	**0.92**	0.63	0.25	**1.00**	0.70
(f)	✓	✓	✓	✓	0.83	**0.75**	**0.42**	0.92	**0.73**

Table 4. Ablation study of KTR and GCM on SIMPLER.

ID	KTR	GCM	Place Bear	Push Button	Clean Table	Scoop Washer	Avg. SR
(a)	✗	✗	0.50	0.45	0.35	0.7	0.50
(b)	✓	✗	0.70	0.45	0.45	0.75	0.59
(c)	✓	✓	**0.75**	**0.6**	**0.55**	**0.85**	**0.69**

Table 5. Ablation study of KTR and GCM on real-world evaluation.

leads to a significant drop in SR from 0.73 to 0.65 and 0.54, respectively. This confirms that both temporal reasoning and memory are essential for long-horizon task completion. Comparing (a)(b)(c), within KTR, the Stage-wise Temporal Structuring (STS) and Keyframe-based Annotation (KBA) each improve SR by +6%/+5%, suggesting the effectiveness of structuring reasoning hierarchically and triggering it at keyframes. Comparing (d)(e)(f), within GCM, Dynamic Reasoning Context (DRC) and Temporal Reasoning Integration (TRI) further improve the SR by +4%/+5%, highlighting the importance of memory buffer and cross-attention retrieval.

5. Conclusion

We present TRM-VLA, a novel Vision-Language-Action framework that integrates temporal-aware reasoning and explicit memory into robotic policy learning. By triggering hierarchical chain-of-thought reasoning only at critical decision points and dynamically maintaining a structured memory of past reasoning traces, TRM-VLA achieves strong performance on long-horizon, memory-intensive manipulation tasks while drastically reducing computational overhead. Extensive experiments across simulated and real-world benchmarks demonstrate that our method sets new state-of-the-art results in both success rate and inference efficiency. We hope TRM-VLA will serve as a foundation for future work on temporally grounded, memory-augmented agents that reason strategically and act efficiently in complex environments.

6. Acknowledgement

This work is supported by the research fund under Grant No. 20242910035 from the Tsinghua University-Jiangsu CRRC Digital Technology Co.,Ltd. Joint Research Center for Data Driven Intelligence of Industry.

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